

Evidence

By a mental mechanism I call naïve empiricism, we have a natural tendency to look for instances that confirm our story and our vision of the world—these instances are always easy to find. Alas, with tools, and fools, anything can be easy to find. You take past instances that corroborate your theories and you treat them as *evidence*. For instance, a diplomat will show you his “accomplishments,” not what he failed to do. Mathematicians will try to convince you that their science is useful to society by pointing out instances where it proved helpful, not those where it was a waste of time, or, worse, those numerous mathematical applications that inflicted a severe cost on society owing to the highly unempirical nature of elegant mathematical theories.

Even in testing a hypothesis, we tend to look for instances where the hypothesis proved true. Of course we can easily find confirmation; all we have to do is look, or have a researcher do it for us. I can *find confirmation* for just about anything, the way a skilled London cabbie can find traffic to increase the fare, even on a holiday.

Some people go further and give me examples of events that we have been able to foresee with some success—indeed there are a few, like landing a man on the moon and the economic growth of the twenty-first century. One can find plenty of “counterevidence” to the points in this book, the best being that newspapers are excellent at predicting movie and theater schedules. Look, I predicted yesterday that the sun would rise today, and it did!

NEGATIVE EMPIRICISM

The good news is that there is a way around this naïve empiricism. I am saying that a series of corroborative facts is not *necessarily* evidence. Seeing white swans does not confirm the nonexistence of black swans. There is an exception, however: I know what statement is wrong, but not necessarily what statement is correct. If I see a black swan I can certify that *all swans are not white!* If I see someone kill, then I can be practically certain that he is a criminal. If I don’t see him kill, I cannot be certain that he is innocent. The same applies to cancer detection: the finding of a single malignant tumor proves that you have cancer, but the absence of such a finding cannot allow you to say with certainty that you are cancer-free.

We can get closer to the truth by negative instances, not by verification! It is misleading to build a general rule from observed facts. Contrary to conventional wisdom, our body of knowledge does not increase from a series of confirmatory observations, like the turkey’s. But there are some things I can remain skeptical about, and others I can safely consider certain. This